PROJECT README

Install Board - Appointment Booking System

A live, double-booking-proof scheduler for vehicle radio installations.

Type	Web application
Organisation	Expert Mobile Communications Ltd.
Year	2026
Role	Designer and developer
Repository	github.com/Masum-jubayel/install-board
Live demo	To be added

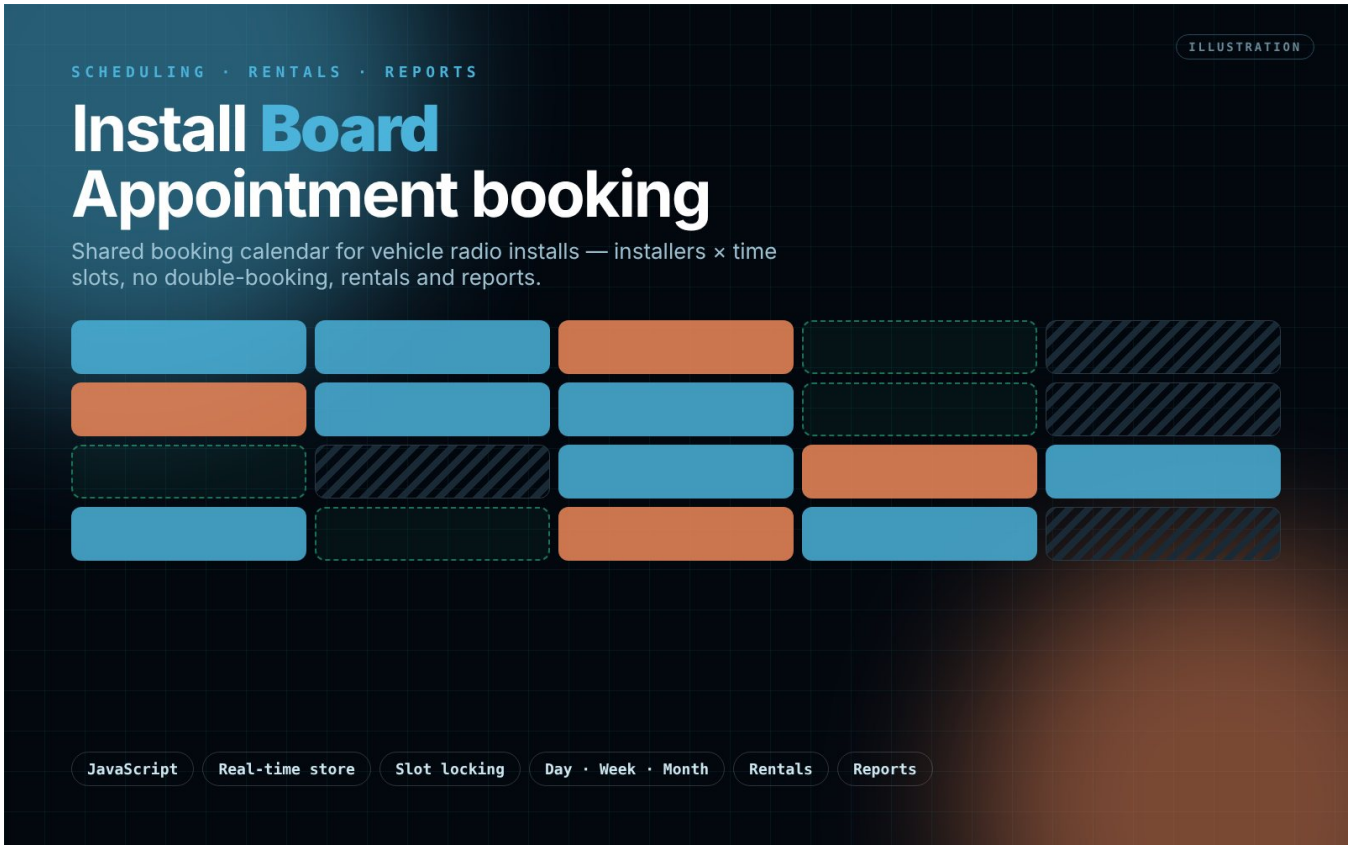


Illustration of the week view. Installer and customer names are not shown.

Contents

1. What the project is 2. Why it was built 3. How it works 4. Features 5. Architecture 6. Technologies 7. Security & privacy 8. Results 9. Next steps

1. What the project is

Install Board is a booking system for vehicle radio installations. Staff book customers into fixed installer slots (08:00, 10:00, 13:00, 15:00), see the whole team in Day, Week or Month view, track rental equipment, block time and print reports. All screens update live.

2. Why it was built

Bookings used to live on paper and in group chats. Double bookings, forgotten rental equipment and no record of who changed what cost installers time and customers trust.

3. How it works, step by step

Step	Stage	What happens
1	**Choose a slot**	Pick an installer and one of the four daily slots in Day, Week or Month view.
2	**Lock**	The app acquires a short-lived lock on that installer and slot, with a time-to-live so abandoned forms release it automatically.
3	**Book**	Customer, vehicle, job type and rental equipment are entered; a ticket number is generated.
4	**Save and release**	The booking is saved while the lock is held, then the lock is released and every open screen updates instantly.
5	**Run the day**	Installers open the daily run sheet; changes are recorded in the history.
6	**Report**	Managers see workload and booking reports and can export them.

4. Features

- Day, Week and Month views across installers
- Slot locks with time-to-live to prevent double booking
- Ticket numbers and searchable bookings list
- Block time for training, holidays or vehicle prep
- Rentals and equipment tracking
- Reports and full change history
- Setup page for installers, slots and rules
- All times in America/Edmonton

5. Architecture

A browser front end talks to a real-time database. A lock collection holds one record per installer + slot with an expiry time. Bookings, rentals, blocks and history are separate collections, so reports are simple queries.

6. Technologies used

Each technology and the job it does in this project:

Technology	What it is used for
JavaScript	The language of the browser; powers the interactive parts of the interface.
TypeScript	JavaScript with static types; catches mistakes before the code runs and makes larger code bases easier to maintain.
HTML5 / CSS3	HTML structures each page; CSS handles layout, colour, typography and responsive design for phones and desktops.
Real-time database	A database that pushes changes to every open screen instantly, so all users see the same live data.
REST APIs	Standard web interfaces used to request and send data between the application and other services.
Concurrency control (slot locks)	Short-lived locks that stop two people from booking the same slot at the same time.
Time-zone handling	All times are stored consistently and displayed in the business's time zone (America/Edmonton).
Responsive UI	The interface adapts to phones, tablets and desktops.

7. Security & privacy

- Only signed-in staff can create or change bookings.
- Every change is recorded with who and when.
- Customer details are not shown in published screenshots.

8. Results

- No double bookings since slot locks went live.
- Front desk and installers share one live board on desktop and phone.
- Every booking has a ticket number and an audit trail.

9. Next steps

- Customer SMS/email reminders
- Calendar (ICS) feed per installer

Author: Masum Jubayel - Network, Security & IT Engineer. Portfolio: masum-jubayel.vercel.app | GitHub: github.com/Masum-jubayel